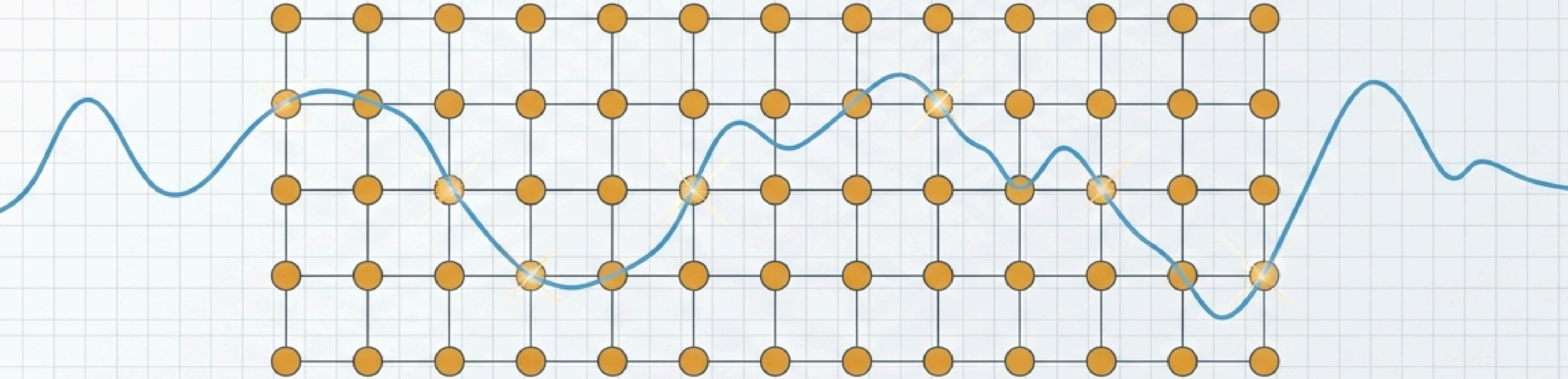
Glucose Twin: Type 2 Diabetes Digital Twin

Proactive, patient-specific forecasting via static and dynamic data fusion.



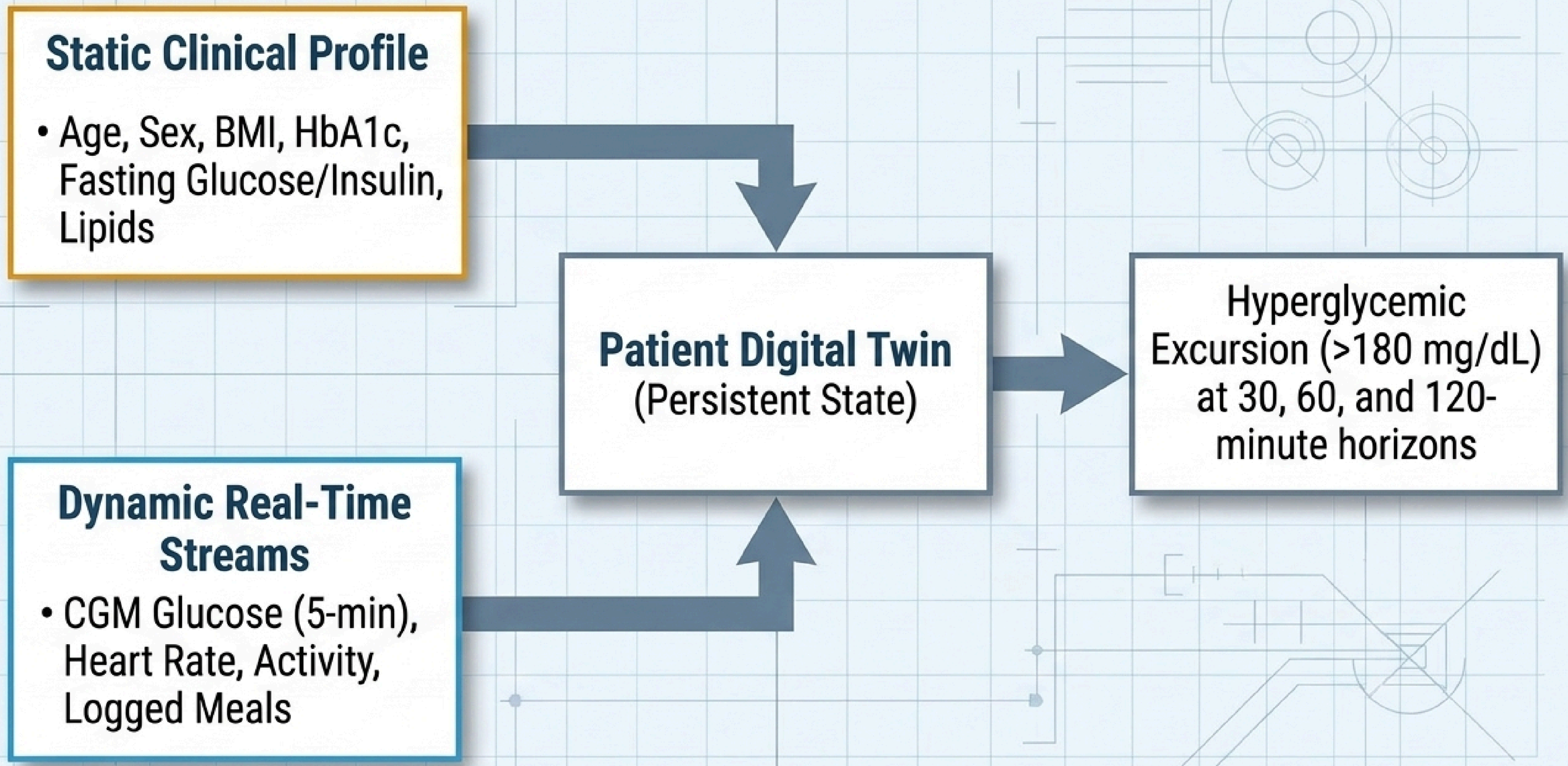
101 Million (11.4%) with Diabetes

136 Million (15.3%) with Prediabetes

Source: ICMR–INDIAB national study, Lancet 2023



CGM data exists, but is currently reviewed retrospectively.
The gap is the absence of a patient-specific, early, explainable warning system.


Static Clinical Profile
• Age, Sex, BMI, HbA1c, Fasting Glucose/Insulin, Lipids
Dynamic Real-Time Streams
• CGM Glucose (5-min), Heart Rate, Activity, Logged Meals
Patient Digital Twin
(Persistent State)
Hyperglycemic Excursion (>180 mg/dL) at 30, 60, and 120-minute horizons

Patient ID	Diagnosis	120-Minute Risk	Status
PT-8092	Type 2	84% ↗	Currently High
PT-1104	Prediabetes	62% ↗	Data Quality Warning
PT-3391	Type 2	12% ↘	Warm-Up

Primary User

Diabetes clinicians & dietitians remotely monitoring patient panels.

Core Goal

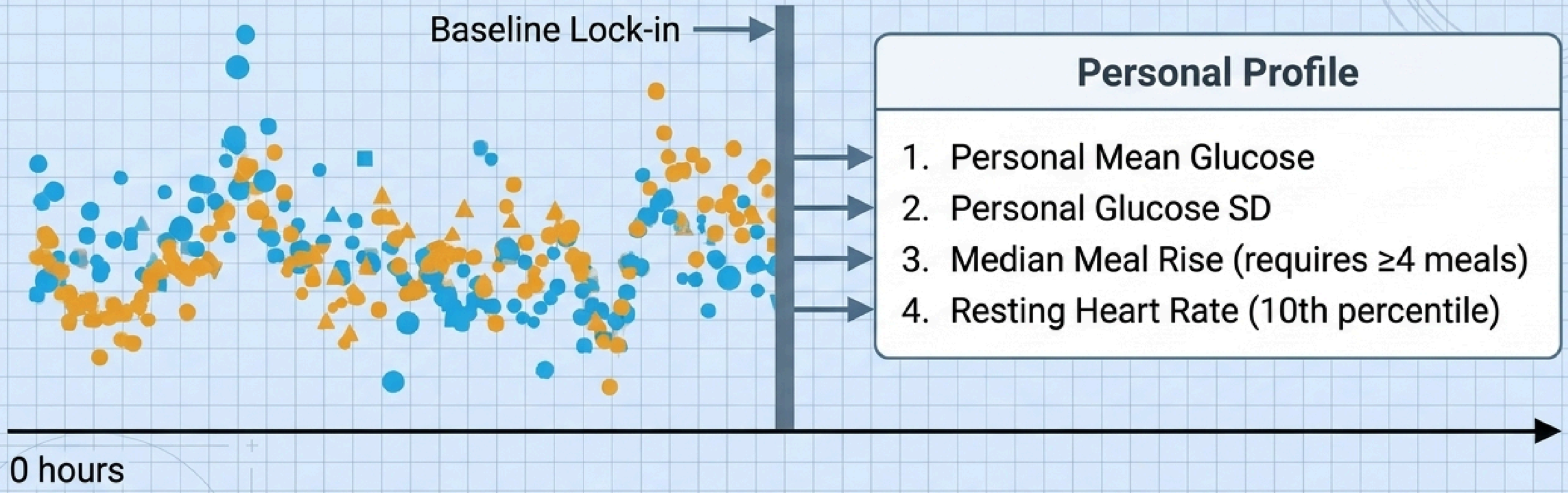
Instantly identify who is heading for a spike and needs immediate attention, minimizing alarm fatigue.

Source	CGMacros (PhysioNet, DOI 10.13026/3z8q-x658)
Cohort	45 Adults (15 Healthy, 16 Prediabetes, 14 Type 2 Diabetes)
Hardware	Dexcom G6 Pro / Libre Pro, Fitbit Sense
Duration	~10 consecutive days per patient, free-living
License	CC BY-NC-SA 4.0 (Non-commercial, share-alike)



LIMITATIONS: Open, real data. US-based cohort only (Santa Barbara). Mostly US-style meals. No simulated data used for modeling. Extrapolations are explicitly labeled.

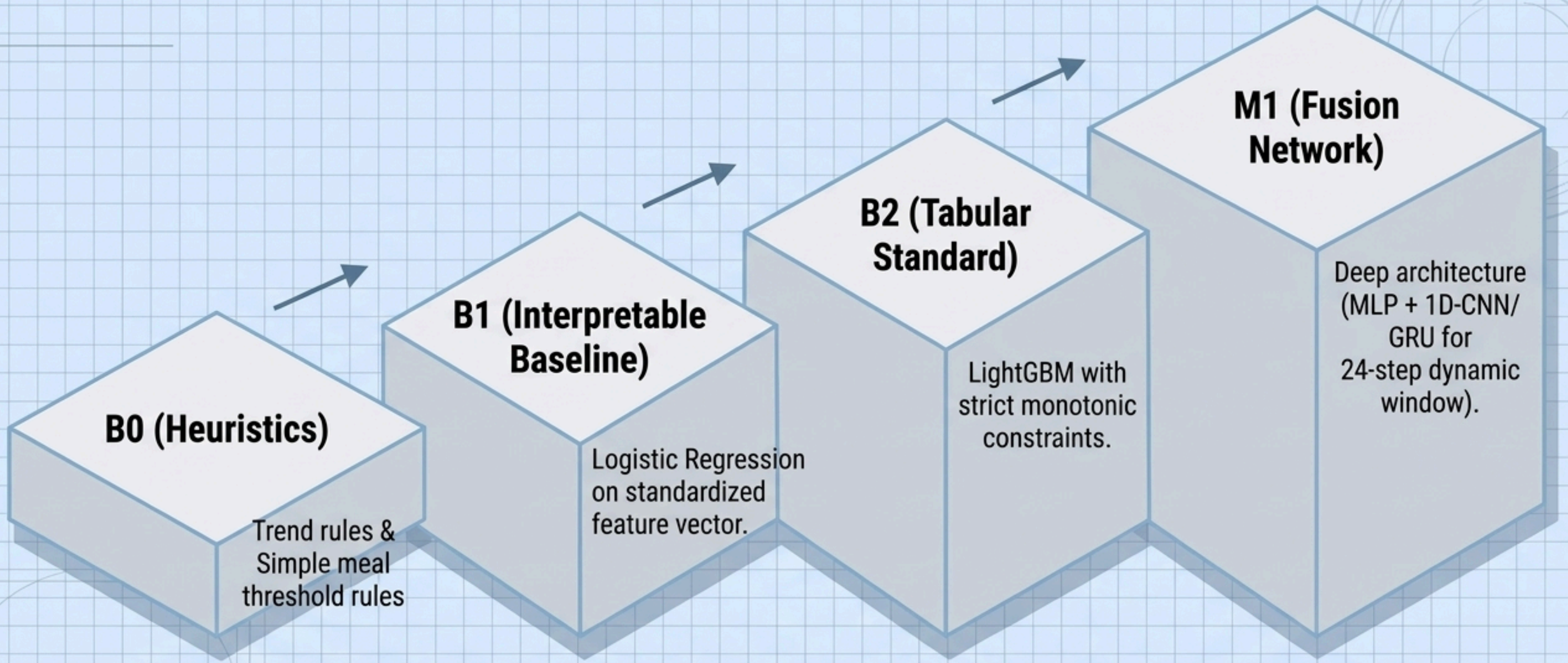
The 48-Hour Warm-Up Phase



Personal Profile

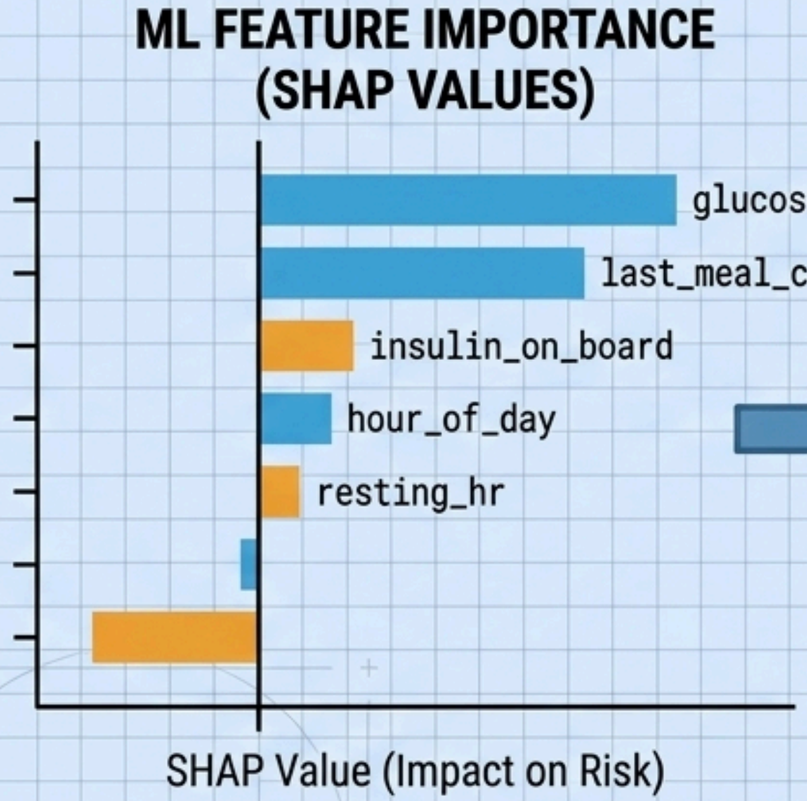
1. Personal Mean Glucose
2. Personal Glucose SD
3. Median Meal Rise (requires ≥4 meals)
4. Resting Heart Rate (10th percentile)

A true twin does not just apply a generic model to new data. It continuously compares real-time telemetry against the patient's own unique biological baseline established during the warm-up phase.



EVALUATION PROTOCOL:
Cross-Validation: 5-fold stratified group CV.
Isolation: No patient appears in both train and test.
Rule: Ship interpretable B2 first; only advance to M1 if out-of-fold PR-AUC proves superiority.

TRANSLATING MACHINE LEARNING TO CLINICAL INSIGHTS



Raw ML Output: Feature Contributions



TRANSLATION ENGINE

Risk is **up** mainly because glucose has risen 38 mg/dL in the last 30 minutes and the last meal had 85 g of carbohydrate.

Actionable Clinical Notification

Every probability score is deconstructed into 3 to 5 plain-language drivers. No black-box scores.

Pre-Meal What-If Scenario

MACRO CONTROLS

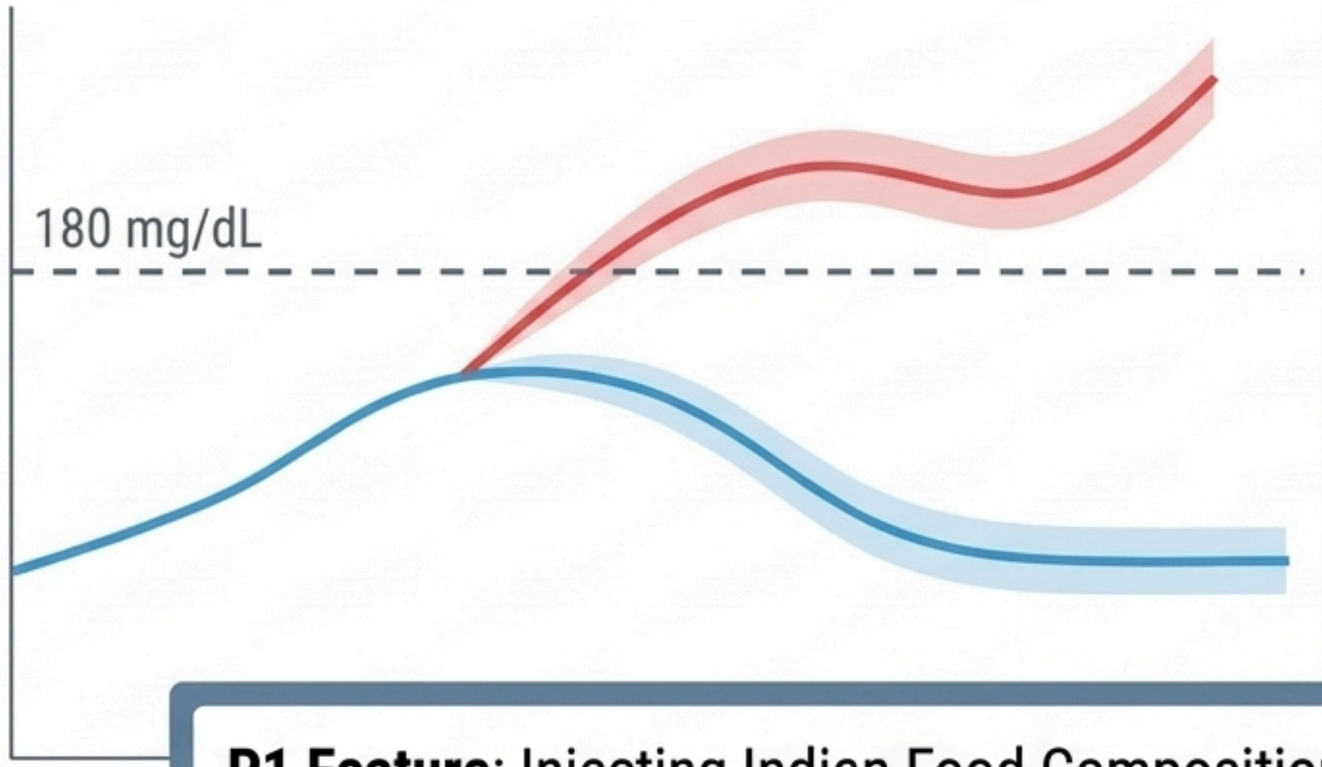
Carbs

Protein

Fat

Fiber

PREDICTED GLUCOSE CURVE



P1 Feature: Injecting Indian Food Composition Tables (NIN-ICMR 2017) as presets (e.g., Dosa, Roti-Dal).

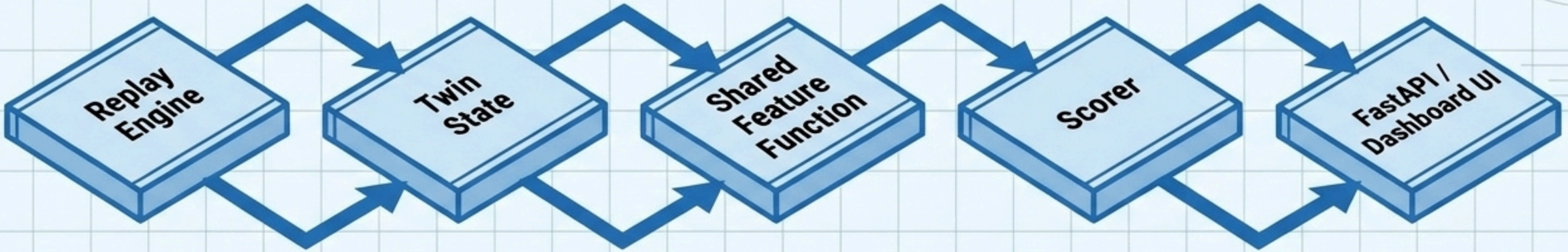
`[ILLUSTRATIVE EXTRAPOLATION: Model trained on US data]`

The Burden of Proof: Ablation Matrix

Variant	Glucose	+ Meals	+ Wearable	+ Static Profile	+ Personal Baseline
A0	✓				
A1	✓	✓			
A2	✓	✓	✓		
A3	✓	✓	✓	✓	
A4 (Full Twin)	✓	✓	✓	✓	✓

TEST T-03: Train/Serve Parity.
Replaying a recorded participant through the twin state must yield the exact same feature vector as the offline batch pipeline. Zero tolerance for drift.

System Architecture & Non-Functional Requirements



NON-FUNCTIONAL REQUIREMENTS:

- **LATENCY:** < 2 seconds from event to updated UI score on laptop CPU.
- **DETERMINISM:** Seeded at 42. Reproducible end-to-end with a single 'make' command.
- **LEAKAGE:** T-04 enforces zero future data leakage; t+1 data cannot influence t predictions.

Measuring Success: Clinical Pragmatism

Primary ML Metrics

PR-AUC
AUROC

Measured at the window level

The Alert Budget

2

Max alerts per patient-day to minimize alarm fatigue

Clinical Utility

>30 Mins

Target Median Lead Time before excursion

Success is not just predictive accuracy; it is maximizing event sensitivity strictly within a tight alarm-fatigue budget, yielding actionable lead time for the clinician.

Clinical Safety & System Limitations



NOT A MEDICAL DEVICE: Advisory forecast from a research prototype only.



NO DIAGNOSIS OR DOSING: System does not simulate treatments or recommend insulin dosing.



DEMOGRAPHIC LIMITS: Proof-of-concept operates on a US cohort. Efficacy on Indian patients is unknown and untested.



MONOTONICITY ENFORCED: System mathematically barred from predicting that increasing carbohydrate intake lowers glucose risk.



CONFIDENCE FLAGS: Predictions are suppressed or flagged ‘low confidence’ if lookback data validity drops below 80%.